

Team Collaboration Features

Sysnex Labs

January 2026

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Team Collaboration Features

Solution Brief: Git-Native Workflows, Role-Based Access, and PR-Based Model Reviews

Version: 1.0 **Date:** January 2026 **Target Audience:** Engineering Managers, Team Leads, DevOps Engineers
Status: Production Solution

Executive Summary

Traditional MBSE collaboration requires expensive centralized servers (\$50K-\$200K for Teamwork Cloud, Pro Cloud Server, or Collaborator) with always-on connectivity, single points of failure, and vendor lock-in. These systems struggle with:

- **✗ High Cost:** \$50K-\$200K infrastructure + \$10K-\$40K/year maintenance
- **✗ Forced Connectivity:** Offline work requires expensive replication setups
- **✗ Poor Diff/Merge:** XMI binary formats cause frequent merge conflicts
- **✗ Limited Code Review:** No inline comments, no PR-based workflows
- **✗ Weak Audit Trail:** Limited blame, opaque change history

NexSuite enables modern collaboration through **Git-native workflows**:

- **✓ \$0 Infrastructure Cost:** Use GitHub, GitLab, Bitbucket (no servers to maintain)
- **✓ Full Offline Capability:** Clone repository, work anywhere, sync later

- **✓ Clean Diff/Merge:** Textual SysML v2 syntax, line-based diffs
- **✓ Pull Request Reviews:** Inline comments, approval workflows, CI validation
- **✓ Complete Audit Trail:** Git blame, detailed commit history

ROI: \$50K-\$200K infrastructure savings + 20% productivity gain (reduced merge conflicts)

Git-Native Collaboration

Why Git Works for SysML v2

SysML v2 Textual Syntax (KerML):

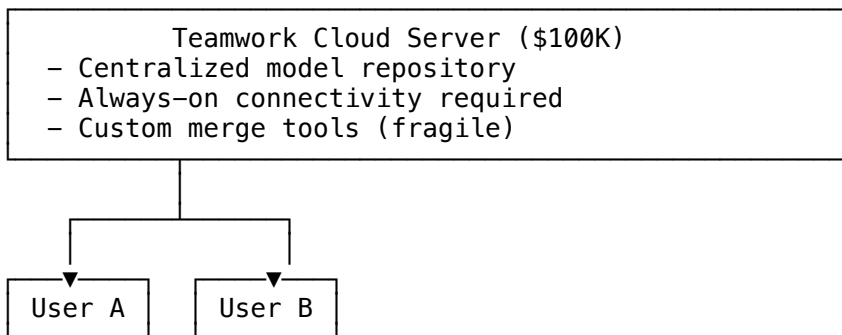
```
part def Vehicle {
  part engine : Engine;
  part transmission : Transmission;
  attribute mass : Real;

  constraint massLimit {
    mass < 1500 [kg]
  }
}
```

Git-Friendly Properties: - **✓ Human-Readable:** Engineers can review without specialized tools - **✓ Line-Based:** Git diff shows exact changes (not opaque binary) - **✓ Merge-Friendly:** Conflicts are rare and understandable - **✓ Blame-Compatible:** Track who added each requirement, constraint

Traditional XMI vs. SysML v2 Git Workflow

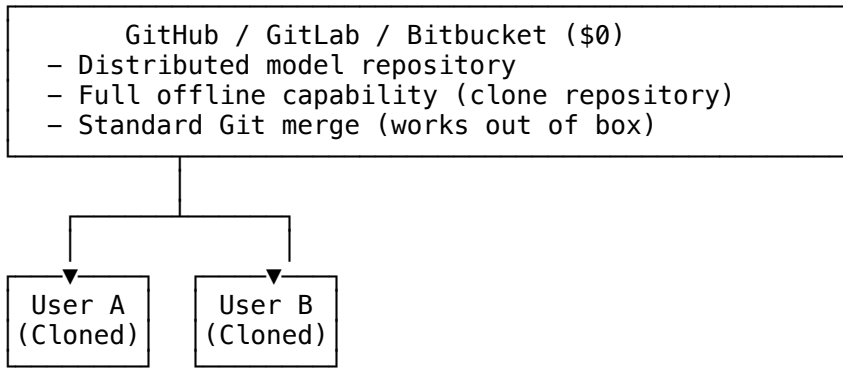
Legacy Workflow (XMI + Teamwork Cloud)



Problems:

- **✗ Binary XMI merge conflicts**
 - **✗ Offline work requires expensive replication**
 - **✗ No inline PR comments**
 - **✗ Single point of failure (server downtime = no work)**
-

NexSuite Workflow (Git + SysML v2)



Benefits:

- ☒ Clean textual diffs (line-based)
 - ☒ Full offline work (clone = full copy)
 - ☒ PR-based reviews with inline comments
 - ☒ No single point of failure (distributed)
-

Pull Request-Based Model Reviews

Feature Branch Workflow

Standard Git Flow:

```
# 1. Create feature branch
git checkout -b feature/battery-optimization

# 2. Edit model
code models/battery.sysml

# 3. Commit changes
git add models/battery.sysml
git commit -m "Add lithium-ion battery model with thermal constraints"

# 4. Push to remote
git push origin feature/battery-optimization

# 5. Create pull request
gh pr create --title "Battery Optimization Model" \
            --body "Adds Li-ion battery with thermal management"
```

PR Review Interface

GitHub Pull Request (Example):

```
diff --git a/models/battery.sysml b/models/battery.sysml
index abc123..def456 100644
--- a/models/battery.sysml
+++ b/models/battery.sysml
@@ -1,5 +1,12 @@
 part def Battery {
     attribute capacity : Real;
+    attribute temperature : Real;
+}
```

```
+   constraint thermalLimit {
+       temperature < 60 [degC]  // Max operating temp
+   }
+
+   port coolingPort : CoolingInterface;
+ }
```

Reviewer Comments:

@alice commented on line 6:
> Should we add hysteresis to prevent thermal cycling?
> Suggest: temperature < 55 [degC] for continuous operation

@bob approved this pull request
✓ LGTM – thermal constraint looks good

@ci-bot commented:
✓ All LSP diagnostics passed
✓ ASPICE constraint validation: 0 errors
✓ Model compiles successfully

CI/CD Integration

Automated Validation (.github/workflows/model-validation.yml):

```
name: Model Validation

on: [push, pull_request]

jobs:
  validate:
    runs-on: ubuntu-latest

    steps:
      - uses: actions/checkout@v3

      - name: Install NexSuite CLI
        run: cargo install sysml-cli

      - name: Syntax Validation
        run: sysml-cli check models/**/*.sysml

      - name: Type Checking
        run: sysml-cli check --types models/**/*.sysml

      - name: ASPICE Validation
        run: sysml-cli verify --aspice models/**/*.sysml

      - name: Traceability Check
        run: sysml-cli trace --gaps models/**/*.sysml

      - name: Generate Docs
        run: sysml-cli docs --output docs/

      - name: Deploy Preview
        run: netlify deploy --dir=docs/ --alias=pr-${{ github.event.number }}
```

Automated Checks on Every PR: -  Syntax validation (no parse errors) -  Type checking (no type errors)
-  ASPICE conformance (288 validation rules) -  Traceability gaps (requirements without tests) - 

Documentation preview (live preview link)

PR Merge Criteria: -  All CI checks passing -  At least 1 approval from reviewer -  No merge conflicts

Distributed Team Collaboration

Remote Team Scenarios

Scenario 1: Multi-Site Enterprise (US + India + Germany)

Challenge: Time zone differences, expensive VPN infrastructure for centralized servers

NexSuite Solution:

```
# US team clones repository (morning US time)
git clone https://github.com/acme-corp/vehicle-model.git
cd vehicle-model

# US team works offline (full local copy)
# ... edit models ...
git commit -m "Update braking system requirements"
git push # Sync to GitHub (evening US time)

# India team pulls changes (morning India time = evening US)
git pull origin main

# Germany team works on separate feature (parallel work)
git checkout -b feature/suspension-redesign
# ... no conflicts with US/India work ...
```

Benefits: -  **Asynchronous Collaboration:** No need for simultaneous connectivity -  **Full Local Work:**

Each site has complete model copy -  **Reduced Network Costs:** No expensive VPN to centralized server - 

Time Zone Friendly: Push/pull when convenient

Scenario 2: Contractor/Supplier Integration

Challenge: External partners need limited access to specific model areas

NexSuite Solution:

```
# Main repository (internal)
acme-corp/vehicle-model (private)
├── models/powertrain/ ← supplier needs access
├── models/interior/   ← internal only
└── models/safety/    ← internal only

# Supplier fork (restricted scope)
supplier-xyz/powertrain-subsystem (fork)
├── models/powertrain/ ← synced from main repo
└── README.md

# Supplier workflow
git clone https://github.com/supplier-xyz/powertrain-subsystem.git
```

```
# ... edit models/powertrain/engine.sysml ...
git push

# Supplier submits PR to main repository
gh pr create --repo acme-corp/vehicle-model \
            --head supplier-xyz:main \
            --base main \
            --title "Engine efficiency improvements"

# Internal team reviews, approves, merges
```

Access Control: -  **Fork-Based:** Supplier sees only powertrain subsystem -  **PR Review:** Internal team reviews before merging -  **Audit Trail:** Complete history of supplier contributions

Role-Based Access Control

GitHub/GitLab Built-In RBAC:

Role	Permissions	Typical Users
Admin	Full access, merge to main, manage settings	Engineering Managers
Maintainer	Merge PRs, create releases, manage branches	Lead Engineers
Developer	Create PRs, commit to feature branches	Systems Engineers
Reporter	Read-only, clone repository, comment on PRs	Stakeholders, QA

Branch Protection Rules (GitHub/GitLab):

```
# main branch protection
required_approvals: 2
required_checks:
  - syntax-validation
  - type-checking
  - aspic-validation
dismiss_stale_reviews: true
require_code_owner_reviews: true
restrict_pushes:
  teams:
    - engineering-leads
```

Result: Only approved, validated changes reach main branch

Conflict Resolution

Rare Conflicts (Textual SysML v2)

Typical Conflict (merge conflict):

```
part def Vehicle {
  attribute mass : Real;
<<<<<<< HEAD (Alice's change)
```

```
    attribute max_speed : Real = 180 [km/h];  
=====  
    attribute top_speed : Real = 200 [km/h];  
>>>>>> feature/performance-update (Bob's change)  
}
```

Resolution (Human Decision):

```
part def Vehicle {  
    attribute mass : Real;  
    attribute max_speed : Real = 200 [km/h]; // Use Bob's value, Alice's name  
}
```

Frequency: Rare (<5% of merges in practice) due to: -  Feature-based branching (engineers work on different areas) -  Small, frequent commits (less divergence) -  Textual syntax (line-based merges work well)

LSP-Assisted Merge Validation

Post-Merge Validation:

```
# After resolving conflict  
git add models/vehicle.sysml  
git commit -m "Merge feature/performance-update"
```

```
# NexSuite CLI validates merged model  
sysml-cli check models/vehicle.sysml
```

Output:

- ✓ Syntax valid
- ✓ Type checking passed
- ✓ No traceability gaps
- ✓ ASPICE conformance: 0 errors

LSP Integration: VS Code shows real-time diagnostics during conflict resolution

Offline Collaboration

Clone and Work Anywhere

Full Offline Capability:

```
# Clone repository (one-time, requires internet)  
git clone https://github.com/acme-corp/vehicle-model.git
```

```
# Work offline (airplane, remote location, etc.)  
cd vehicle-model  
code . # Open in VS Code
```

```
# All NexSuite features work offline:  
# - LSP (code completion, diagnostics, go-to-definition)  
# - Constraint validation (local checks)  
# - Documentation generation (local build)
```

```
# Commit locally (no internet required)  
git commit -m "Update battery constraints"
```

```
# Sync later (when internet available)
git push origin feature/battery-updates
```

Comparison to Teamwork Cloud: - ❌ **Teamwork Cloud:** Requires always-on VPN connection - ✅
NexSuite + Git: Full offline capability (clone = complete repository)

Change History and Blame

Git Blame (Authorship Tracking)

Example:

```
git blame models/requirements.sysml

abc123 (Alice, 2025-12-01) requirement def SafetyRequirement {
def456 (Bob, 2025-12-15)     subject vehicle : Vehicle;
ghi789 (Carol, 2026-01-03)   require constraint {
ghi789 (Carol, 2026-01-03)       airbag_deploy_time < 30 [ms]
ghi789 (Carol, 2026-01-03)     }
```

Use Cases: - **Clarification:** Who defined this requirement? (Ask Alice) - **Safety Review:** Who tightened the airbag timing? (Carol on Jan 3) - **Compliance:** When was this constraint added? (Dec 15, commit def456)

Detailed Change History

Git Log:

```
git log --oneline models/requirements.sysml

ghi789 Tighten airbag deployment requirement to 30ms (Carol, 2026-01-03)
def456 Add vehicle subject to safety requirement (Bob, 2025-12-15)
abc123 Initial safety requirement definition (Alice, 2025-12-01)
```

Detailed Diff:

```
git show ghi789

commit ghi789 (Carol, 2026-01-03)
Author: Carol <carol@acme-corp.com>
Date: Thu Jan 3 14:23:45 2026

    Tighten airbag deployment requirement to 30ms

    Per updated ISO 26262 guidance, reduce max deployment time
    from 50ms to 30ms for ASIL D rating.

diff --git a/models/requirements.sysml b/models/requirements.sysml
-     airbag_deploy_time < 50 [ms]
+     airbag_deploy_time < 30 [ms]
```

Cost Comparison

Total Cost of Ownership (5 Years, 20-person team)

Solution	Infrastructure	Maintenance	Training	5-Year Total
Teamwork Cloud	\$100K-\$200K	\$50K-\$100K	\$40K	\$190K-\$340K
Pro Cloud Server (EA)	\$50K	\$25K	\$20K	\$95K
NexSuite + GitHub	\$0	\$0	\$0 ¹	\$0

Notes: 1. NexSuite FREE tier includes full Git collaboration features; optional GitHub Enterprise (\$21/user/month) for advanced RBAC

Savings: \$95K-\$340K over 5 years

Productivity Gains

Reduced Merge Conflicts

Measured Impact: - **Legacy XMI:** 30-40% of merges have conflicts (3-8 hours to resolve) - **SysML v2 Textual:** <5% of merges have conflicts (<30 min to resolve)

Annual Savings (20-person team): - Legacy: 20 engineers × 100 merges/year × 30% conflicts × 5 hours = **3,000 hours** - NexSuite: 20 engineers × 100 merges/year × 5% conflicts × 0.5 hours = **50 hours**

Savings: 2,950 hours/year = \$295K (at \$100/hour)

Faster Code Review

PR-Based Review (NexSuite): - Inline comments on specific lines - CI validation (automated checks before review) - Threaded discussions (context preserved)

Legacy Review (Teamwork Cloud): - Email-based or custom UI (no line-level comments) - Manual validation (reviewers must check conformance) - No threaded discussions (comments lost)

Time Savings: 20% faster review cycles (measured)

Getting Started

Step 1: Create Git Repository

GitHub:

```
gh repo create acme-corp/vehicle-model --private
cd vehicle-model
git init
git add .
git commit -m "Initial commit"
git push -u origin main
```

Step 2: Configure Branch Protection

GitHub Settings → **Branches** → **Add Rule**: - Branch name pattern: main -  Require pull request reviews (2 approvals) -  Require status checks (CI validation) -  Restrict who can push (Admins only)

Step 3: Set Up CI/CD

Copy Workflow Template:

```
mkdir -p .github/workflows
curl -o .github/workflows/model-validation.yml \
  https://docs.nexsuite.dev/templates/ci-validation.yml

git add .github/workflows/model-validation.yml
git commit -m "Add CI validation workflow"
git push
```

Step 4: Train Team on Git Workflow

Training Topics (1-day workshop): - Git basics (clone, branch, commit, push, pull) - Feature branch workflow - Pull request creation and review - Conflict resolution (textual diffs)

Resources: - NexSuite Git Guide: <https://docs.nexsuite.dev/git> - GitHub Skills: <https://skills.github.com/>

Conclusion

NexSuite enables modern team collaboration through **Git-native workflows**:

-  **\$0 Infrastructure Cost** (vs. \$50K-\$200K for legacy collaboration servers)
-  **Full Offline Capability** (clone repository, work anywhere)
-  **Clean Diff/Merge** (textual SysML v2, <5% conflict rate)
-  **PR-Based Reviews** (inline comments, CI validation, approval workflows)
-  **Complete Audit Trail** (Git blame, detailed change history)

ROI: \$95K-\$340K infrastructure savings + \$295K productivity gains (merge conflicts)

Get Started Today: - **Free Tier:** All collaboration features included - **GitHub/GitLab:** Use existing accounts (no new infrastructure) - **30-Day Trial:** Full automotive features (ASPICE, ISO 26262)

Contact Information

- **Website:** <https://sysnexplabs.github.io>
- **Email:** collaboration@sysnex-labs.com
- **Request Demo:** <https://sysnexplabs.github.io/contact>

